# IFCSR: Inference-Free Fidelity-Realism Control for One-Step Diffusion-based Real-World Image Super-Resolution

Jonghee Back* Jongju Kim* Jeong-Uk Kim* Eunjin Kim Minyong Jeon†
BLUEDOT Inc.
{jongheeback2020,0012siri2100}@gmail.com {jayden.kim,eunice.kim,mickey.jeon}@blue-dot.io

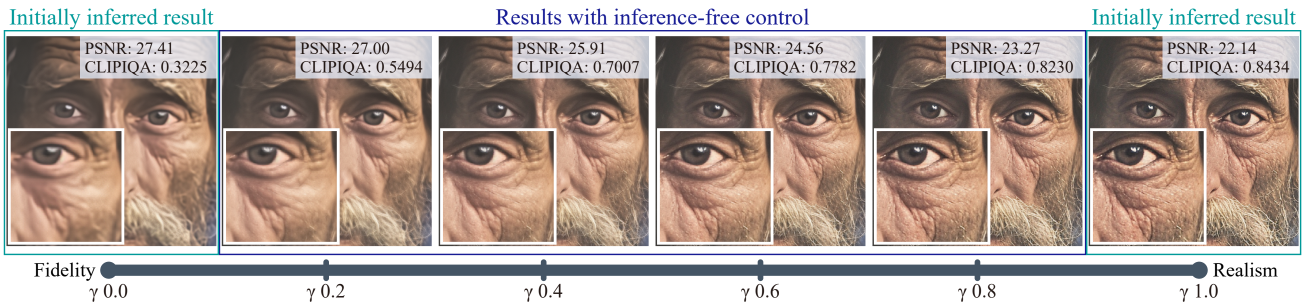



Figure 1. Visual results of our controllable framework for real-world image super-resolution. Users can flexibly explore the fidelity-realism trade-off via a single controllable parameter $\gamma$ with no extra inference (e.g., the four middle images), once fidelity- and realism-specific images (i.e., the two outermost images) are initially inferred. As $\gamma$ increases, the fidelity metric (PSNR) gradually decreases while the realism metric (CLIPIQA [33]) improves, demonstrating effective fidelity-realism control without further network evaluation.

## Abstract

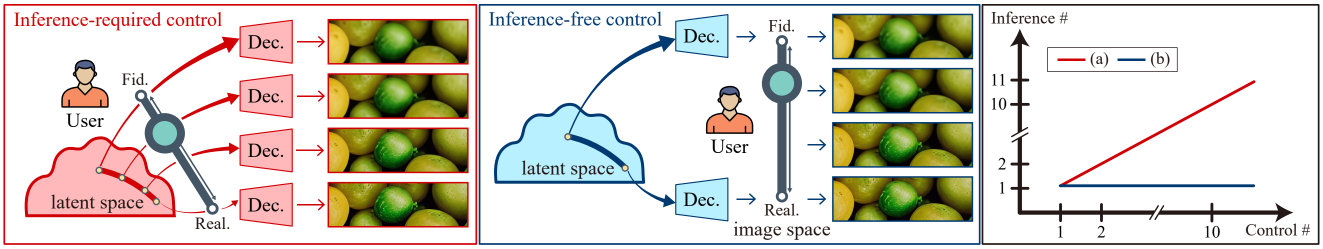
*Diffusion models have recently achieved remarkable success in real-world image super-resolution (ISR), typically balancing a trade-off between fidelity (i.e., similarity to HR images) and realism (i.e., perceptual naturalness). To better account for subjective preferences in image quality, controllable diffusion-based methods have been explored, allowing personalized adjustment of this trade-off via tunable parameters. While existing controllable methods have shown effective control, they typically operate in the latent space and require repeated network inference during adjustment, eventually limiting their practicality. In this paper, we propose IFCSR, a simple yet practical approach for one-step diffusion-based real-world ISR that enables inference-free control between fidelity and realism. The key idea is to design a controllable model that adjusts the fidelity-realism trade-off in the image space, rather than in the latent space. Such an image-space control allows users to seamlessly adjust the trade-off without extra inference after an initial inference of fidelity- and realism-specific images. We further introduce a two-stage training scheme and specialized losses that encourage the controllable space to span a broad spectrum of fidelity and realism. Our method achieves quality competitive with state-of-the-art models while providing a practical advantage through inference-free control.*

*Equal contribution.
†Corresponding author.

## 1. Introduction

Image super-resolution (ISR) [7, 8, 18, 19, 24, 32], which aims to restore high-resolution (HR) images from low-resolution (LR) images, remains a long-standing challenge. In general, the quality of super-resolved images is evaluated using two main criteria: *fidelity*, which measures how closely an output resembles its HR counterpart, and *realism*, which indicates how perceptually natural and photorealistic an output appears to viewers [2]. While fidelity is improved by minimizing reconstruction losses (e.g., MSE loss), realism is commonly enhanced by leveraging generative priors (e.g., GAN [12]) to better align with natural image distributions [5, 20, 35, 36, 53]. In recent years, diffusion models [14, 30] have emerged as powerful tools for generating realistic visuals, and leveraging pre-trained text-to-image (T2I) models (e.g., Stable Diffusion (SD) [28]) as generative priors has become a promising direction to achieve realistic ISR [4, 9, 10, 34, 42, 43, 46, 47, 49].

While these approaches aim to produce high-quality im-




(a) Control mechanism (previous approaches) (b) Control mechanism (our approach) (c) Number of inferences per control

Figure 2. Comparison of control mechanisms in controllable diffusion-based ISR methods. (a) Previous approaches [21, 31, 44] control fidelity and realism in the latent space, requiring multiple forward passes of their neural networks (e.g., the VAE decoder and the SD UNet, depending on the design) to generate controlled images. (b) Our method performs this control directly in the image space, with no extra inference after generating fidelity- and realism-specific images. (c) While the number of inference (i.e., single forward pass) in previous methods grows linearly with each control, ours remains constant, offering a clear practical advantage for real-world use.

ages balanced between fidelity and realism, controllable diffusion-based ISR methods [21, 31, 44, 56], which allow users to adjust the fidelity-realism trade-off according to their subjective preferences, have recently been explored. Their main idea is to introduce controllable parameters and design a diffusion-based architecture that enables the model to produce HR images specialized for either fidelity or realism depending on the parameters. PiSA-SR [31] employs two separate models to modulate the fidelity-realism trade-off, and other approaches [21, 44, 56] employ a timestep as a controllable parameter for the trade-off.

Despite their effective controllability over the fidelity-realism spectrum, they typically rely on manipulating the trade-off in the latent space, which requires a full forward pass through neural networks (e.g., the VAE decoder and the SD UNet, depending on the design) to generate the final image (see Fig. 2 (a)). In practice, users need to iteratively conduct numerous inferences to match their preferences. These repeated inferences hamper practicality in real-world applications because the number of inferences increases linearly with controllable parameter adjustments, as illustrated by the red-colored line in Fig. 2 (c). Even with the acceleration of diffusion-based ISR methods from multiple steps to few or one, they remain computationally expensive, thus making inference-required control impractical.

To address the aforementioned limitation, we propose IFCSR, a practical framework for one-step diffusion-based ISR that enables *inference-free control* between fidelity and realism. Our key idea is to formulate a controllable model between fidelity- and realism-specific networks *in the image space rather than the latent space*, as shown in Fig. 2 (b). Once an initial inference for two specialized networks is complete, this controllable model linearly blends their distinct outputs through a tunable parameter, allowing users to flexibly navigate the fidelity-realism spectrum without any further network evaluation (see the blue-colored line in Fig. 2 (c)). Furthermore, to encourage fidelity- and realism-specific networks to specialize in either fidelity or realism, we adopt a two-stage training scheme that optimizes the networks separately and design specialized losses tailored for the respective training stages.

As illustrated in Fig. 1, varying our adjustable parameter $\gamma$ effectively controls the trade-off between fidelity and realism (i.e., higher fidelity metric (PSNR) on the left and higher realism metric (CLIPIQA [33]) on the right). Notably, the resulting images during control (see the four middle results in Fig. 1) are generated without further network inference. We demonstrate that IFCSR produces high-quality results competitive with state-of-the-art controllable methods while offering a clear practical advantage, i.e., inference-free control. Moreover, a balanced parameter setting achieves quality comparable to recent ISR approaches.

Our contributions are summarized as below:

- We propose a practical diffusion-based framework for real-world ISR that enables fidelity-realism adjustment in the image space, offering inference-free control.
- We introduce a two-stage scheme to sequentially train fidelity- and realism-specific networks, aligning the controllable space with the fidelity-realism spectrum.
- We design a feature-space loss with depth-dependent weighting that guides the specialized networks toward fidelity or realism, enabling a wider controllable spectrum.

## 2. Related Work

Deep learning has been widely applied to ISR tasks, achieving notable improvements over traditional methods [39]. While early works [7, 20, 35] were trained on LR-HR image pairs synthesized via simple degradation, recent studies [27, 36, 53] have adopted complex degradation models to improve robustness in diverse real-world scenarios, known as real-world ISR. We focus on diffusion-based real-world ISR, including controllable methods.

**Diffusion-based real-world ISR** Recently, the success of diffusion models [14, 30] has spurred interest in applying them to ISR, especially by leveraging pre-trained T2I models as generative priors [34, 43, 46, 49]. Despite their impressive quality, diffusion-based methods require multi-step sampling, causing non-trivial computational cost and limiting their practical use in real-world applications.

To mitigate this limitation, recent studies have focused on distilling the multi-step diffusion process into few- or one-step models, enabling more efficient diffusion-based ISR [4, 9, 10, 37, 42, 47, 48, 50, 51]. ResShift [50] progressively shifted the LR-HR residual to reduce sampling steps to as few as 15, and SinSR [37] further shortened this to a single step via consistency-preserving distillation. OSEDiff [42] employed variational score distillation [40] in the latent space to condense SD into one step, and AdcSR [4] and TVT [47] extended OSEDiff to further reduce its computational cost. TSD-SR [9] proposed target score distillation with a distribution-aware sampling module, which was advanced by TinySR [10] to enhance model efficiency. InvSR [51] proposed diffusion inversion for ISR by introducing a noise predictor that takes the LR image as input, enabling efficient arbitrary-step sampling. CTMSR [48] achieved distillation-free one-step ISR via consistency training and enhanced realism through distribution trajectory matching. While these diffusion-based methods significantly improve efficiency, they offer limited flexibility for users to manipulate fidelity and realism according to individual preferences.

**Controllable diffusion-based real-world ISR** To improve the flexibility of ISR, early diffusion-based methods provided initial capabilities for user-facing control between fidelity and realism, e.g., controllable feature wrapping module [34] and adjustable noise scheduling [46]. With the growing demand for finer control, recent studies [21, 31, 44, 56] have proposed specialized methods to enhance controllability. PiSA-SR [31] trained two distinct low-rank adapters (LoRAs) [15] for fidelity and realism in the SD UNet, enabling latent interpolation between the two. OFTSR [56] and CTSR [21] distilled models to achieve fidelity-realism control using a timestep within the diffusion. RCOD [44] controlled the generative strength by assigning a timestep according to input degradation.

Despite their effectiveness, these approaches perform control in the latent space or within the diffusion model, which requires repeated inference for each adjustment. In contrast, we aim to directly manipulate fidelity and realism in the image space without extra inference during control.

## 3. Method

We first formulate our controllable model in the image space, which allows us to adjust fidelity and realism without requiring additional inference (Sec. 3.1). We then introduce a two-stage training scheme that aligns the controllable space of our model with the fidelity-realism spectrum (Sec. 3.2). Lastly, we propose fidelity- and realism-specific losses that enable our model to be more specialized for fidelity and realism, respectively (Sec. 3.3).

### 3.1. Formulation

Our goal is to formulate an ISR model that enables users to adjust the fidelity-realism trade-off without extra inference. To this end, we introduce two specialized one-step diffusion networks $f_\theta$ and $f_\phi$ that take as input a low-quality (LQ) image $\mathbf{x}_L$, each of which aims to predict a high-quality (HQ) image emphasizing fidelity and realism, respectively. We then define our controllable model for the final HQ image $\hat{\mathbf{x}}_H$ as a linear combination of the two outputs, $\hat{\mathbf{x}}_H^{\text{fid}}$ and $\hat{\mathbf{x}}_H^{\text{real}}$:

$$\begin{aligned} \hat{\mathbf{x}}_H &= (1-\gamma) f_\theta(\mathbf{x}_L) + \gamma f_\phi(\mathbf{x}_L) \\ &= (1-\gamma)\hat{\mathbf{x}}_H^{\text{fid}} + \gamma \hat{\mathbf{x}}_H^{\text{real}}, \end{aligned} \tag{1}$$

where $\gamma \in [0, 1]$ is a controllable parameter. Intuitively, Eq. (1) produces images with high fidelity but low realism as $\gamma$ is close to zero, whereas it generates low-fidelity but high-realism images as $\gamma$ approaches one. Note that linear combination in the image space enables inference-free control between fidelity and realism via the parameter $\gamma$, once $\hat{\mathbf{x}}_H^{\text{fid}}$ and $\hat{\mathbf{x}}_H^{\text{real}}$ are obtained.

Regarding the fidelity- and realism-specific networks, we follow previous one-step diffusion-based architectures [31, 42], which utilize an LQ image as the starting point for diffusion process. Specifically, our diffusion networks encode the LQ image into a latent vector using the VAE encoder, perform the LQ-to-HQ latent transformation via the SD UNet, and decode the predicted latent vector into the image space using the VAE decoder. We adopt the residual learning [31] for the LQ-to-HQ transformation. Each network has its own trainable low-rank adapter (LoRA) [15] attached to the pre-trained SD UNet, parameterized by $\theta$ for fidelity and $\phi$ for realism. We denote them as $\text{LoRA}_\theta$ and $\text{LoRA}_\phi$, respectively.

To optimize the fidelity- and realism-specific networks, we can minimize the discrepancy between the output $\hat{\mathbf{x}}_H$ and the ground truth image $\mathbf{x}_H$ over the entire range of $\gamma$, using the following objective function:

$$[\hat{\theta}, \hat{\phi}] = \arg\min_{\theta,\phi} \mathbb{E}_{\gamma\sim U(0,1)}[\mathcal{L}(\hat{\mathbf{x}}_H, \mathbf{x}_H)], \tag{2}$$

where $\mathcal{L}$ denotes a loss function. During training, the controllable parameter $\gamma$ is sampled from a uniform distribution $U(0, 1)$ at each step, while during inference, it can be tuned according to user preferences.

However, due to the lack of an explicit mechanism to specialize $f_\theta$ and $f_\phi$ for fidelity and realism, this objective function alone does not guarantee the alignment of our controllable space with the fidelity-realism trade-off, thus limiting controllability between the two. In the following two

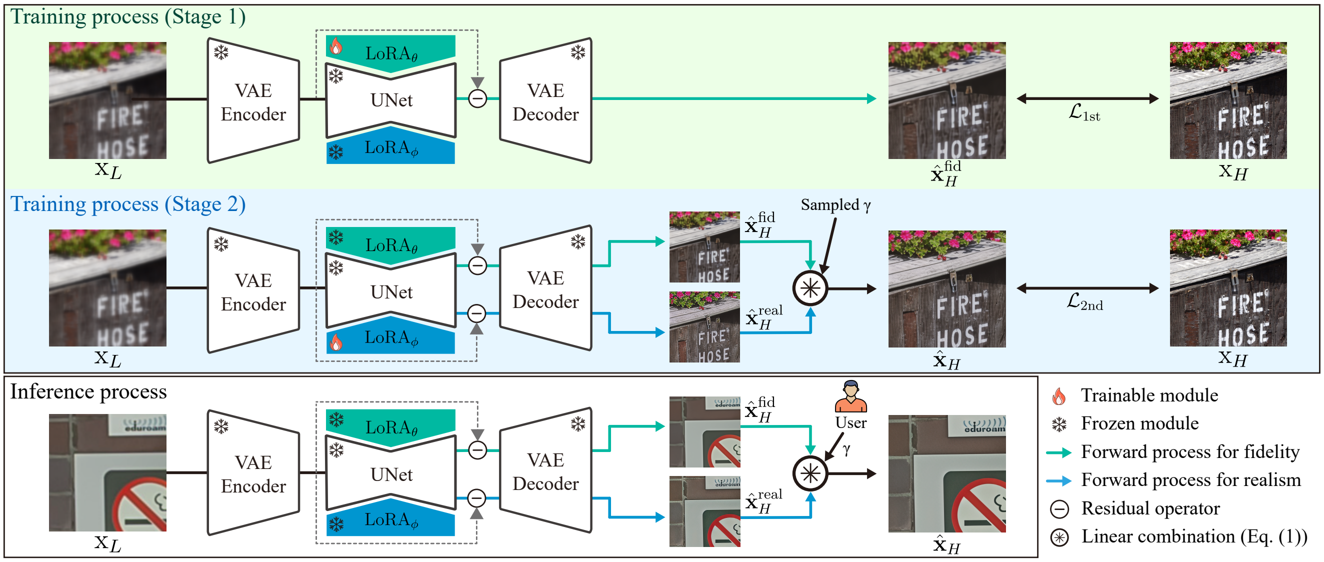



Figure 3. Overview of the training and inference processes of our approach. In Stage 1 (top row), we train the fidelity-specific network with a trainable LoRA$_\theta$ using our specialized loss $\mathcal{L}_{1st}$, which focuses on producing a high-fidelity image $\hat{\mathbf{x}}_H^{\text{fid}}$. In Stage 2 (middle row), we optimize the realism-specific network with another trainable LoRA$_\phi$. Both $\hat{\mathbf{x}}_H^{\text{fid}}$ and $\hat{\mathbf{x}}_H^{\text{real}}$ are inferred and linearly combined with a sampled $\gamma$. Then, the combined output $\hat{\mathbf{x}}_H$ is supervised against the HQ target $\mathbf{x}_H$ via the realism-specific loss $\mathcal{L}_{2nd}$ while keeping $\theta$ frozen. In the inference stage (bottom row), after producing $\hat{\mathbf{x}}_H^{\text{fid}}$ and $\hat{\mathbf{x}}_H^{\text{real}}$, users can control fidelity-realism trade-off by tuning $\gamma$ with no inference.

sections, we will discuss how to effectively train the fidelity- and realism-specific diffusion networks.

### 3.2. Two-Stage Training Scheme

To achieve controllability between fidelity and realism, we introduce a two-stage training scheme that enables each diffusion network to specialize in one of the two objectives, as shown in the top and middle rows of Fig. 3. This design decouples the optimization objectives of fidelity and realism, allowing each network to focus exclusively on its respective goal. In the first stage, we train the fidelity-specific network $f_\theta$ only and then optimize the realism-specific network $f_\phi$ while freezing the fidelity-specific one.

**Stage 1. Training the fidelity-specific network** We first fine-tune $f_\theta$ with a trainable LoRA$_\theta$, as shown in the top row of Fig. 3. At this stage, since $f_\phi$ is not trained yet, we set $\gamma$ to zero in our controllable model (i.e., Eq. (1)) and minimize the loss between the predicted and ground truth images, which eventually indicates the objective function:

$$\hat{\theta} = \arg\min_{\theta} \mathcal{L}_{1st}(\hat{\mathbf{x}}_H^{\text{fid}}, \mathbf{x}_H), \tag{3}$$

where $\mathcal{L}_{1st}$ is a loss function for Stage 1. Since improving fidelity is relatively straightforward compared to realism (e.g., minimizing reconstruction losses such as MSE), it is a natural choice to train the fidelity-specific network first, rather than the realism-specific one.

**Stage 2. Training the realism-specific network** In the next stage, we train $f_\phi$ with a trainable LoRA$_\phi$ as shown in the middle row of Fig. 3. During this stage, $\hat{\mathbf{x}}_H^{\text{fid}}$ and $\hat{\mathbf{x}}_H^{\text{real}}$ are inferred from $f_\theta$ and $f_\phi$ for our controllable model while freezing $\theta$ to ensure that $\phi$ is updated only, as defined by the following objective function:

$$\hat{\phi} = \arg\min_{\phi} \mathbb{E}_{\gamma\sim U(0,1)}[\mathcal{L}_{2nd}(\hat{\mathbf{x}}_H, \mathbf{x}_H)], \tag{4}$$

where $\mathcal{L}_{2nd}$ is a loss function for Stage 2. Here, $\gamma$ is sampled from a uniform distribution at each training step. This training configuration encourages the realism-specific network to strengthen realism and ultimately aligns the learned controllable space along the fidelity-realism spectrum.

The remaining technical question is which loss function (i.e., $\mathcal{L}_{1st}$ and $\mathcal{L}_{2nd}$ in Eqs. (3) and (4)) to adopt for training the fidelity- and realism-specific networks at each stage. Following prior works [9, 42], the combination of LPIPS [55] and MSE losses serves as a straightforward baseline. However, since this loss is not explicitly tailored to either fidelity or realism, it offers limited controllability between the two aspects, potentially reducing its practical utility. The design of our specialized losses for the two-stage training is discussed in the following section.

### 3.3. Fidelity- and Realism-Specific Losses

We extend the LPIPS loss [55], which measures distances between feature representations from a pre-trained network

| Datasets | Methods | PSNR ↑ | SSIM ↑ | LPIPS ↓ | DISTS ↓ | FID ↓ | NIQE ↓ | MUSIQ ↑ | MANIQA ↑ | CLIPIQA ↑ |
|---|---|---|---|---|---|---|---|---|---|---|
| RealSR | StableSR-s200 | 24.62 | 0.7036 | 0.3070 | 0.2151 | 126.25 | 5.8180 | 65.92 | 0.6278 | 0.6312 |
| | DiffBIR-s50 | 24.83 | 0.6532 | 0.3596 | 0.2398 | 128.92 | 5.7617 | 68.96 | 0.6490 | **0.7096** |
| | SeeSR-s50 | 25.15 | 0.7211 | 0.3007 | 0.2224 | 125.35 | 5.3969 | 69.82 | 0.6451 | 0.6697 |
| | PASD-s20 | **26.64** | **0.7661** | 0.2866 | **0.2032** | 124.65 | 5.9746 | 60.00 | 0.5366 | 0.4854 |
| | ResShift-s15 | 26.43 | 0.7584 | 0.3139 | 0.2424 | 153.32 | 6.9134 | 60.05 | 0.5351 | 0.5452 |
| | SinSR-s1 | 26.20 | 0.7375 | 0.3069 | 0.2321 | 137.49 | 6.0355 | 61.03 | 0.5411 | 0.6209 |
| | OSEDiff-s1 | 25.15 | 0.7341 | 0.2921 | 0.2128 | 123.52 | 5.6391 | 69.09 | 0.6331 | 0.6685 |
| | AdcSR-s1 | 25.47 | 0.7301 | 0.2885 | 0.2129 | 118.42 | 5.3480 | 69.90 | 0.6353 | 0.6731 |
| | TSD-SR-s1 | 24.81 | 0.7172 | 0.2743 | 0.2105 | 114.51 | **5.1258** | **71.18** | 0.6346 | **0.7158** |
| | InvSR-s1 | 24.45 | 0.7138 | 0.2852 | 0.2115 | 137.59 | 5.6476 | 68.33 | **0.6596** | 0.6786 |
| | TVT-s1 | 25.81 | 0.7596 | **0.2597** | 0.2061 | **109.87** | 5.9197 | 69.89 | 0.6232 | 0.6885 |
| | CTMSR-s1 | 25.99 | 0.7543 | 0.2899 | 0.2206 | 136.66 | 5.5467 | 64.36 | 0.5273 | 0.6391 |
| | PiSA-SR-s1 | 25.50 | 0.7418 | **0.2672** | **0.2045** | 124.20 | 5.5009 | 70.15 | 0.6552 | 0.6701 |
| | Ours-s1 ($\gamma = 0.0$) | **28.00** | **0.7975** | 0.2921 | 0.2187 | 138.88 | 7.9015 | 47.64 | 0.4025 | 0.3337 |
| | Ours-s1 ($\gamma = 0.5$) | 25.14 | 0.7055 | 0.2866 | 0.2141 | **110.60** | 5.4313 | 69.68 | 0.6293 | 0.6522 |
| | Ours-s1 ($\gamma = 1.0$) | 22.10 | 0.6075 | 0.3502 | 0.2367 | 115.31 | **4.9010** | **70.29** | **0.6643** | 0.7011 |
| DRealSR | StableSR-s200 | 27.96 | 0.7465 | 0.3329 | 0.2295 | 154.00 | 6.5206 | 59.37 | 0.5633 | 0.6242 |
| | DiffBIR-s50 | 25.73 | 0.6109 | 0.4795 | 0.2882 | 168.35 | 6.4500 | **67.66** | **0.6249** | 0.7055 |
| | SeeSR-s50 | 28.07 | 0.7684 | 0.3173 | 0.2314 | 147.36 | 6.4047 | 65.07 | 0.6051 | 0.6908 |
| | PASD-s20 | **29.20** | **0.7961** | 0.3175 | 0.2217 | 142.52 | 7.6835 | 50.26 | 0.4663 | 0.5027 |
| | ResShift-s15 | 28.71 | 0.7893 | 0.3496 | 0.2538 | 179.51 | 7.7935 | 52.67 | 0.4743 | 0.5415 |
| | SinSR-s1 | 28.18 | 0.7530 | 0.3539 | 0.2497 | 179.03 | 6.6590 | 57.38 | 0.5016 | 0.6604 |
| | OSEDiff-s1 | 27.92 | 0.7835 | 0.2967 | **0.2162** | 135.59 | 6.4423 | 64.67 | 0.5895 | 0.6958 |
| | AdcSR-s1 | 28.10 | 0.7726 | 0.3046 | 0.2200 | 134.05 | 6.4444 | 66.27 | 0.5915 | 0.7048 |
| | TSD-SR-s1 | 27.77 | 0.7559 | 0.2967 | **0.2136** | 135.24 | **5.9213** | 66.61 | 0.5874 | **0.7356** |
| | InvSR-s1 | 26.40 | 0.7152 | 0.3516 | 0.2437 | 168.13 | 5.9342 | 65.81 | **0.6275** | 0.7149 |
| | TVT-s1 | 28.27 | 0.7899 | **0.2900** | 0.2205 | 134.22 | 7.0345 | 65.56 | 0.5776 | 0.7219 |
| | CTMSR-s1 | 28.66 | 0.7836 | 0.3238 | 0.2361 | 161.46 | 6.2074 | 59.93 | 0.4868 | 0.6521 |
| | PiSA-SR-s1 | 28.32 | 0.7804 | **0.2960** | 0.2169 | **130.46** | 6.1774 | 66.11 | 0.6145 | 0.6975 |
| | Ours-s1 ($\gamma = 0.0$) | **30.86** | **0.8472** | 0.3179 | 0.2340 | 156.39 | 8.5957 | 42.32 | 0.3380 | 0.3471 |
| | Ours-s1 ($\gamma = 0.5$) | 27.60 | 0.7371 | 0.3278 | 0.2361 | **133.84** | 6.0570 | 66.57 | 0.5940 | 0.7149 |
| | Ours-s1 ($\gamma = 1.0$) | 24.63 | 0.6377 | 0.3957 | 0.2587 | 144.00 | **5.7044** | **67.14** | 0.6229 | **0.7446** |

Table 1. Quantitative comparison with state-of-the-art diffusion-based ISR methods on the RealSR [3] and DRealSR [41] datasets. 's' stands for the number of diffusion steps. We highlight the best and the second-best results with **blue** and **green**, respectively.

(e.g., VGG [29]), to further specialize two diffusion networks in their respective aspects. Motivated by the observation [52] that early layers capture low-level structures (e.g., edges and colors) while deeper layers extract high-level semantics (e.g., class-specific information), our key idea is to apply depth-dependent weighting to per-layer feature distances in the LPIPS. By assigning monotonically increasing or decreasing weights across layers, we enable each network to focus on its intended objective.

Following [55], let $\mathcal{D}(\hat{\mathbf{x}}_H, \mathbf{x}_H, l)$ denote the L2 distance between the unit-normalized features of $\hat{\mathbf{x}}_H$ and $\mathbf{x}_H$ at layer $l \in \{1, 2, \cdots, L\}$ of a pre-trained network (e.g., VGG network) with $L$ layers. $\mathcal{D}(\hat{\mathbf{x}}_H, \mathbf{x}_H, 0)$ indicates the L2 distance of unit-normalized $\hat{\mathbf{x}}_H$ and $\mathbf{x}_H$. We then introduce a new loss as the weighted average of the feature distances:

$$\mathcal{L}_s(\hat{\mathbf{x}}_H, \mathbf{x}_H; w) = \frac{1}{\sum_{j=0}^{L} w_j} \sum_{l=0}^{L} w_l \mathcal{D}(\hat{\mathbf{x}}_H, \mathbf{x}_H, l), \quad (5)$$

where $w_l$ is the weight at layer $l$. For fidelity-specific training (Stage 1), we emphasize shallow features by assigning monotonically decreasing weights $w_l^{\text{dec}} = k^{L-l}$. For realism-specific training (Stage 2), we highlight deep features via monotonically increasing weights $w_l^{\text{inc}} = k^{l-1}$. Here, $k \geq 1$ is a hyperparameter that controls the growth or decay rate. Note that, when $k$ is set to one, the loss behaves like LPIPS loss with MSE on unit-normalized inputs.

In summary, our fidelity- and realism-specific losses for the two-stage training are defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{1st}}(\hat{\mathbf{x}}_H^{\text{fid}}, \mathbf{x}_H) &= \mathcal{L}_s(\hat{\mathbf{x}}_H^{\text{fid}}, \mathbf{x}_H; w^{\text{dec}}), \\ \mathcal{L}_{\text{2nd}}(\hat{\mathbf{x}}_H, \mathbf{x}_H) &= \mathcal{L}_s(\hat{\mathbf{x}}_H, \mathbf{x}_H; w^{\text{inc}}). \end{aligned} \quad (6)$$

### 3.4. Inference Stage

During inference, as illustrated in the bottom row of Fig. 3, the trained fidelity- and realism-specific networks generate their respective outputs, i.e., $\hat{\mathbf{x}}_H^{\text{fid}}$ and $\hat{\mathbf{x}}_H^{\text{real}}$, from a given LQ input $\mathbf{x}_L$. These outputs are then linearly combined in the image space using the controllable parameter $\gamma$, as defined in Eq. (1), to produce the final image $\hat{\mathbf{x}}_H$. It is worth noting that our method allows inference-free control once the initial inference is complete, enabling users to flexibly explore the fidelity-realism spectrum by simply adjusting $\gamma$.

## 4. Experiments

### 4.1. Experimental Settings

**Datasets** This paper focuses on the 4× ISR task. We train our model using the FFHQ [16] (first 10,000 images) and LSDIR [22] (84,991 images) datasets, consistent with previous ISR methods [31, 42, 43]. We randomly crop

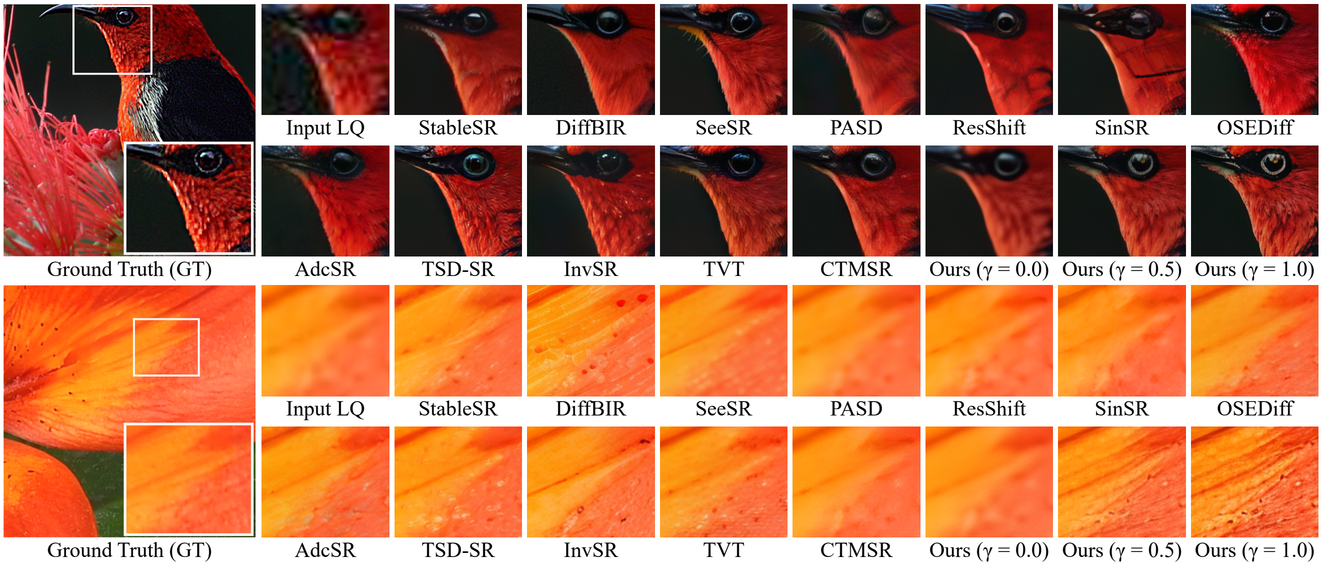



Figure 4. Visual comparison with recent diffusion-based ISR methods, tested on 0853_PCH_00019 (top) and P1171051 (bottom) images.

$512 \times 512$ patches from the dataset and apply the degradation introduced in Real-ESRGAN [36] to produce pairs of synthetic LQ and HQ images for training. For evaluation, we utilize three benchmark datasets: RealSR [3] and DRealSR [41] for real-world scenarios, and DIV2K-val [1] for synthetic scenarios, following the evaluation settings in previous works [31, 42, 43]. To test our method, we upscale LR images from $128 \times 128$ to $512 \times 512$ resolutions using bicubic interpolation, which serves as our LQ inputs.

**Compared methods and evaluation metrics** We have compared our method with state-of-the-art diffusion-based ISR methods (StableSR [34], DiffBIR [25], SeeSR [43], PASD [46], ResShift [50], SinSR [37], OSEDiff [42], AdcSR [4], TSD-SR [9], InvSR [51], TVT [47], and CTMSR [48]) and controllable approaches (StableSR, PASD, and PiSA-SR [31]). We have tested these approaches using their publicly released codes and pre-trained models. Due to space constraints, additional comparisons with non-diffusion-based approaches [5, 23, 36, 53] are provided in the supplementary report.

We evaluate the quantitative quality of the methods using reference-based metrics (PSNR, SSIM [38], LPIPS [55], and DISTS [6]) and no-reference metrics (NIQE [54], MUSIQ [17], MANIQA [45], and CLIPIQA [33]). Following the perception-distortion discussion by [2], we treat the former as fidelity metrics and the latter as realism metrics. Moreover, we measure FID [13], which compares the feature distributions between predicted and GT images.

**Implementation details** We adopt Stable Diffusion v2-1-base [28] as the pre-trained T2I model and set the LoRA rank to 4 for all LoRAs in our method. For simplicity, we use a null prompt for the T2I model. In both the first and second stages, we train our method using a batch size of 4 and the AdamW [26] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and learning rate 5e-5. Unless otherwise specified, we set $k$ to 7 and employ a pre-trained VGG network [29] for computing the fidelity- and realism-specific losses. The training process took about 24 hours for each stage (i.e., total 48 hours) on a single NVIDIA H100 GPU.

## 4.2. Comparisons with State-of-the-Art Methods

In this section, we compare our method with recent diffusion-based ISR methods, including controllable ones. During inference, we use $\gamma \in \{0.0, 0.5, 1.0\}$ for visual and numerical comparisons, and a denser set ($\gamma \in \{0.0, 0.2, \cdots, 1.0\}$) to show the fidelity-realism spectrum.

**Comparisons with diffusion-based methods** We provide quantitative comparison results between diffusion-based methods and our proposed method in Table 1. Our method with $\gamma = 0.0$ achieves high reference-based metrics (e.g., highest PSNR and SSIM scores), indicating superior fidelity compared to the others. Conversely, setting $\gamma = 1.0$ results in high no-reference metrics (e.g., best NIQE and CLIPIQA on DRealSR), indicating a stronger focus on perceptual realism. Ours with $\gamma = 0.5$ balances fidelity and realism, generating results that are competitive with other recent methods across all metrics.

Fig. 4 presents a visual comparison among diffusion-based ISR methods, including ours. Our results with $\gamma = 0.0$ prioritize fidelity over realism, generating clean yet perceptually over-smoothed images compared with other ap-

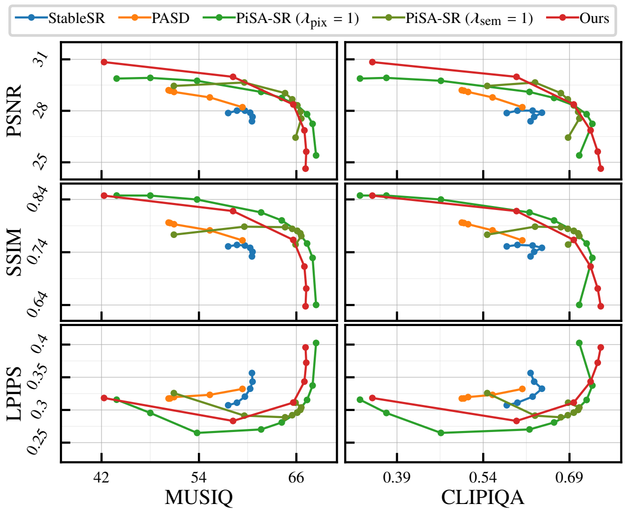
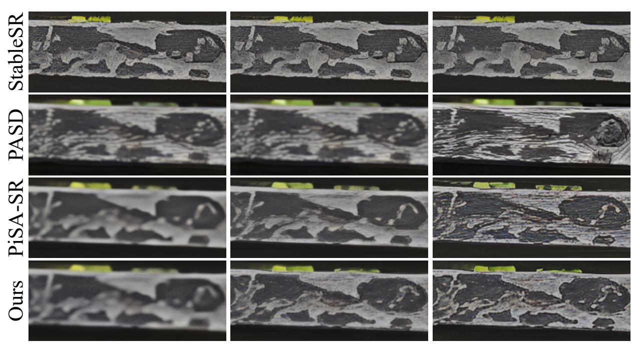



Figure 5. Comparison of controllable ISR methods on the DRealSR dataset, illustrating the fidelity-realism spectrum. The y-axis represents PSNR, SSIM, and LPIPS, and the x-axis represents MUSIQ and CLIPIQA. StableSR and PASD are evaluated using $\omega \in [0.0, 1.0]$ and $n \in [100, 900]$, respectively. For PiSA-SR, two curves are presented: one varying $\lambda_{sem} \in [0.0, 1.5]$ with $\lambda_{pix} = 1$, and the other varying $\lambda_{pix} \in [0.0, 1.2]$ with $\lambda_{sem} = 1$.




(a) Param. setting (fidelity-oriented) (b) Param. setting (central) (c) Param. setting (realism-oriented)

Figure 6. Visual comparison of fidelity-to-realism transition for controllable ISR methods on the NIKON_005 image from the RealSR dataset. We chose their fidelity- and realism-oriented parameters ((a) and (c)) according to the metrics evaluated on the RealSR and central value (b) of their parameters, i.e., StableSR ($\omega \in \{0.8, 0.6, 0.4\}$), PASD ($n \in \{900, 700, 500\}$), and PiSA-SR ($\lambda_{pix} = 1.0, \lambda_{sem} \in \{0.2, 0.8, 1.5\}$).

proaches. Setting $\gamma = 1.0$ improves perceptual quality with sharper details and richer textures (see the bird feathers (top) and the petal (bottom) in Fig. 4) than competing methods. When $\gamma = 0.5$, it produces visually balanced outputs that preserve fidelity while enriching details. More quantitative and visual comparisons (e.g., DIV2K-val results) can be found in the supplementary report.

| $\gamma$ | PSNR ↑ | SSIM ↑ | LPIPS ↓ | NIQE ↓ | MUSIQ ↑ | CLIPIQA ↑ |
|---|---|---|---|---|---|---|
| 0.0 | **28.00** | **0.7975** | 0.2921 | **7.8936** | **47.64** | **0.3329** |
| 0.2 | 27.34 | 0.7764 | **0.2540** | 6.2632 | 61.58 | 0.5093 |
| 0.4 | 25.89 | 0.7297 | 0.2726 | 5.6356 | 68.73 | 0.6244 |
| 0.6 | 24.41 | 0.6825 | 0.3008 | 5.2439 | 70.12 | 0.6711 |
| 0.8 | 23.13 | 0.6416 | 0.3275 | 4.9833 | **70.36** | 0.6913 |
| 1.0 | **22.10** | **0.6075** | **0.3501** | **4.9002** | 70.29 | **0.7008** |
| 0.0 | **30.84** | **0.8472** | 0.3183 | **8.6472** | **42.31** | **0.3472** |
| 0.2 | 29.98 | 0.8179 | **0.2832** | 6.8617 | 58.20 | 0.5979 |
| 0.4 | 28.37 | 0.7633 | 0.3113 | 6.1694 | 65.62 | 0.6974 |
| 0.6 | 26.86 | 0.7129 | 0.3434 | 5.8771 | 66.99 | 0.7267 |
| 0.8 | 25.62 | 0.6712 | 0.3723 | 5.7476 | **67.22** | 0.7391 |
| 1.0 | **24.63** | **0.6376** | **0.3957** | **5.6919** | 67.13 | **0.7444** |

Table 2. Numerical results of our method with varying $\gamma$, evaluated on RealSR (top) and DRealSR (bottom). The best and worst scores across $\gamma$ are highlighted in **blue** and **brown**, respectively.

**Comparisons with controllable methods** Fig. 5 shows the fidelity-realism spectrum of controllable diffusion-based methods evaluated under varying controllable parameters. The x-axis presents no-reference metrics (MUSIQ and CLIPIQA), and the y-axis denotes reference-based metrics (PSNR, SSIM, and LPIPS). In the PSNR/SSIM and MUSIQ/CLIPIQA plots (top and middle rows), both PiSA-SR and our method, designed for fidelity-realism control, exhibit clearer and wider spectra than StableSR and PASD, and our method achieves quality comparable to PiSA-SR. However, PiSA-SR and our method show slight non-monotonic behavior in some cases, e.g., fidelity-oriented settings of the LPIPS-based spectrum (bottom row). For our method, this non-monotonic trend is primarily amplified by our loss (see Sec. 4.3 and the supplementary report for details).

Furthermore, a visual comparison among the controllable models is presented in Fig. 6. In contrast to StableSR and PASD, both PiSA-SR and our approach generate progressively richer textures on the wooden board as the parameter setting shifts toward realism from fidelity. Notably, our approach enables a clear visual transition through $\gamma$, producing competitive quality to PiSA-SR while offering higher practicality via inference-free control.

### 4.3. Discussions

**Controllability analysis of our method** To verify the controllability of our method, we evaluate how evaluation metrics vary with different values of $\gamma$, as shown in Table 2. As $\gamma$ increases from zero to one, image-space fidelity metrics (PSNR and SSIM) gradually decrease, whereas realism metrics consistently improve, except for MUSIQ at $\gamma = 1.0$, which remains nearly the same as at $\gamma = 0.8$. While the overall trend of image-space fidelity metrics and realism metrics demonstrates the controllability of our method, it is shown that the feature-space fidelity metric (e.g., LPIPS) exhibits an inconsistent trade-off across $\gamma$. We observe that using our feature-space loss with depth-dependent weighting amplifies this inconsistency (see the supplementary re-

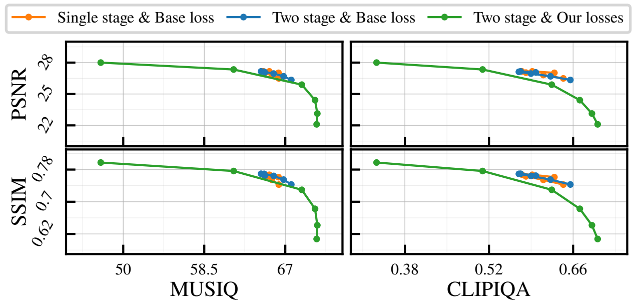



Figure 7. Ablation study demonstrating the effectiveness of our two-stage training scheme and specialized losses on the RealSR dataset. We plot the fidelity-realism spectrum of three settings: single- and two-stage training using the baseline loss (LPIPS + MSE), and two-stage training using our losses.

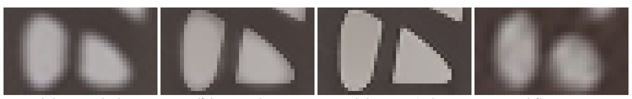



Figure 8. Failure case of our method where fidelity- and realism-specific images ((a) and (c)) differ significantly in structure.

port for results using different values of $k$). Nonetheless, except for this metric, our approach enables smooth transitions along the fidelity-realism spectrum via $\gamma$.

**Effectiveness of our two-stage training scheme and specialized losses** We conduct three experiments to validate the effectiveness of our two-stage training scheme and specialized losses as shown in Fig. 7: single-stage training (i.e., Eq. (2)) using the baseline loss, and two-stage training (i.e., Eqs. (3) and (4)) using the same baseline loss and our specialized losses. We use LPIPS combined with MSE as the baseline loss. As illustrated by the orange-colored lines, single-stage training leads to a misalignment with the fidelity-realism spectrum due to the absence of specialization for fidelity and realism. While our two-stage training begins to establish this alignment (see the blue-colored lines), the resulting spectrum remains narrow since the baseline loss is not explicitly tailored for either fidelity or realism. Incorporating our specialized losses into the two-stage training significantly widens the spectrum by further encouraging each stage to specialize in its respective goal, resulting in more effective representations in the controllable spectrum, as shown by the green-colored lines.

**Ablation studies under different settings** We conducted ablation studies to examine the effect of our design choices, including specialized losses with different $k$ and an alternative two-stage training scheme. For detailed results and discussions, please refer to the supplementary report.

| | Step counts | Inference time (s) | Parameter counts (B) |
|---|---|---|---|
| StableSR | 200 | 12.39 | 1.55 |
| DiffBIR | 50 | 6.15 | 1.68 |
| SeeSR | 50 | 6.28 | 2.51 |
| PASD | 20 | 2.74 | 2.31 |
| ResShift | 15 | 0.91 | 0.17 |
| SinSR | 1 | 0.20 | 0.17 |
| OSEDiff | 1 | 0.17 | 1.76 |
| AdcSR | 1 | 0.08 | 0.46 |
| TSD-SR | 1 | 0.17 | 2.21 |
| InvSR | 1 | 0.17 | 0.46 |
| TVT | 1 | 0.49 | 1.73 |
| CTMSR | 1 | 0.12 | 0.17 |
| PiSA-SR | 1 | 0.20 | 1.29 |
| Ours | 1 | 0.35 | 1.30 |

Table 3. Model efficiency comparison across diffusion-based ISR models. Inference time is reported in seconds on an NVIDIA RTX 3090 GPU with FP16 precision and a batch size of 1, and parameter counts are presented in billions. TVT is evaluated with FP32 precision as FP16 precision is not available in the released codes.

**Limitations and future work** While we adopt an image-space linear combination for inference-free fidelity-realism control, it can cause semi-transparent artifacts in a blended image when the fidelity- and realism-specific images differ significantly in structure (e.g., edges), as shown in Fig. 8. Moreover, our method prioritizes effectiveness and practicality, and thus its efficiency is not fully optimized. As presented in Table 3, our method is more efficient than multi-step methods but slightly slower than other single-step ones due to generating both fidelity- and realism-specific images. It also retains a relatively large number of parameters. It would be interesting to explore extensive ways to improve efficiency, e.g., network pruning [4, 10, 11]. Another potential direction is to adapt score distillation methods for ISR [9, 31, 42] within our controllable framework, potentially enhancing realism further. A straightforward incorporation of an OSEDiff-style VSD loss shows promising gains in no-reference metrics, as reported in the supplementary report. These results motivate a more effective incorporation as a promising direction for future work.

## 5. Conclusion

We propose IFCSR, a controllable one-step diffusion framework for real-world image super-resolution, focusing on enhancing practicality through inference-free control. Unlike existing controllable techniques operating in the latent space, we introduce a controllable model that adjusts the fidelity-realism trade-off in the image space, enabling flexible control without any additional inference. Built upon this controllable formulation, the proposed two-stage training scheme and specialized losses effectively align the controllable space along the fidelity-realism spectrum. Extensive experiments demonstrate that IFCSR achieves quality competitive with state-of-the-art methods while offering the practical advantage of inference-free adjustment.

## Acknowledgments

We appreciate the anonymous reviewers for their constructive comments. This research was supported by the High-Performance Computing Support Project (G2025-0795), funded by the Government of the Republic of Korea (Ministry of Science and ICT).

## References

[1] Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2017. 6

[2] Yochai Blau and Tomer Michaeli. The perception-distortion tradeoff. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 1, 6

[3] Jianrui Cai, Hui Zeng, Hongwei Yong, Zisheng Cao, and Lei Zhang. Toward real-world single image super-resolution: A new benchmark and a new model. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. 5, 6

[4] Bin Chen, Gehui Li, Rongyuan Wu, Xindong Zhang, Jie Chen, Jian Zhang, and Lei Zhang. Adversarial diffusion compression for real-world image super-resolution. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 28208–28220, 2025. 1, 3, 6, 8

[5] Chaofeng Chen, Xinyu Shi, Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, and Shihui Guo. Real-world blind super-resolution via feature matching with implicit high-resolution priors. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 1329–1338, 2022. 1, 6

[6] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P. Simoncelli. Image quality assessment: Unifying structure and texture similarity. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(5):2567–2581, 2022. 6

[7] Chao Dong, Chen Change Loy, Kaiming He, and Xiaoou Tang. Learning a deep convolutional network for image super-resolution. In *European Conference on Computer Vision (ECCV)*, pages 184–199. Springer, 2014. 1, 2

[8] Chao Dong, Chen Change Loy, and Xiaoou Tang. Accelerating the super-resolution convolutional neural network. In *European Conference on Computer Vision (ECCV)*, pages 391–407. Springer, 2016. 1

[9] Linwei Dong, Qingnan Fan, Yihong Guo, Zhonghao Wang, Qi Zhang, Jinwei Chen, Yawei Luo, and Changqing Zou. Tsd-sr: One-step diffusion with target score distillation for real-world image super-resolution. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 23174–23184, 2025. 1, 3, 4, 6, 8

[10] Linwei Dong, Qingnan Fan, Yuhang Yu, Qi Zhang, Jinwei Chen, Yawei Luo, and Changqing Zou. Tinysr: Pruning diffusion for real-world image super-resolution. *arXiv preprint arXiv:2508.17434*, 2025. 1, 3, 8

[11] Gongfan Fang, Xinyin Ma, and Xinchao Wang. Structural pruning for diffusion models. In *Advances in Neural Information Processing Systems*, pages 16716–16728, 2023. 8

[12] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, 2014. 1

[13] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, 2017. 6

[14] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, pages 6840–6851, 2020. 1, 2

[15] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. 3

[16] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 5

[17] Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5148–5157, 2021. 6

[18] Jiwon Kim, Jung Kwon Lee, and Kyoung Mu Lee. Deeply-recursive convolutional network for image super-resolution. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 1

[19] Jiwon Kim, Jung Kwon Lee, and Kyoung Mu Lee. Accurate image super-resolution using very deep convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 1

[20] Christian Ledig, Lucas Theis, Ferenc Huszar, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, and Wenzhe Shi. Photo-realistic single image super-resolution using a generative adversarial network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 1, 2

[21] Runyi Li, Bin Chen, Jian Zhang, and Radu Timofte. Ctsr: Controllable fidelity-realness trade-off distillation for real-world image super resolution. *arXiv preprint arXiv:2503.14272*, 2025. 2, 3

[22] Yawei Li, Kai Zhang, Jingyun Liang, Jiezhang Cao, Ce Liu, Rui Gong, Yulun Zhang, Hao Tang, Yun Liu, Denis Demandolx, Rakesh Ranjan, Radu Timofte, and Luc Van Gool. Lsdir: A large scale dataset for image restoration. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 1775–1787, 2023. 5

[23] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. Swinir: Image restoration using swin transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 1833–1844, 2021. 6

[24] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single

image super-resolution. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2017. 1

[25] Xinqi Lin, Jingwen He, Ziyan Chen, Zhaoyang Lyu, Bo Dai, Fanghua Yu, Yu Qiao, Wanli Ouyang, and Chao Dong. Diffbir: Toward blind image restoration with generative diffusion prior. In *European Conference on Computer Vision (ECCV)*, pages 430–448. Springer, 2024. 6

[26] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. 6

[27] Ziwei Luo, Fredrik K. Gustafsson, Zheng Zhao, Jens Sjölund, and Thomas B. Schön. Photo-realistic image restoration in the wild with controlled vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 6641–6651, 2024. 2

[28] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022. 1, 6

[29] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014. 5, 6

[30] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. 1, 2

[31] Lingchen Sun, Rongyuan Wu, Zhiyuan Ma, Shuaizheng Liu, Qiaosi Yi, and Lei Zhang. Pixel-level and semantic-level adjustable super-resolution: A dual-lora approach. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 2333–2343, 2025. 2, 3, 5, 6, 8

[32] Ying Tai, Jian Yang, and Xiaoming Liu. Image super-resolution via deep recursive residual network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 1

[33] Jianyi Wang, Kelvin CK Chan, and Chen Change Loy. Exploring clip for assessing the look and feel of images. In *AAAI*, 2023. 1, 2, 6

[34] Jianyi Wang, Zongsheng Yue, Shangchen Zhou, Kelvin CK Chan, and Chen Change Loy. Exploiting diffusion prior for real-world image super-resolution. *International Journal of Computer Vision*, 132(12):5929–5949, 2024. 1, 3, 6

[35] Xintao Wang, Ke Yu, Shixiang Wu, Jinjin Gu, Yihao Liu, Chao Dong, Yu Qiao, and Chen Change Loy. Esrgan: Enhanced super-resolution generative adversarial networks. In *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, 2018. 1, 2

[36] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 1905–1914, 2021. 1, 2, 6

[37] Yufei Wang, Wenhan Yang, Xinyuan Chen, Yaohui Wang, Lanqing Guo, Lap-Pui Chau, Ziwei Liu, Yu Qiao, Alex C. Kot, and Bihan Wen. Sinsr: Diffusion-based image super-resolution in a single step. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25796–25805, 2024. 3, 6

[38] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4): 600–612, 2004. 6

[39] Zhihao Wang, Jian Chen, and Steven C. H. Hoi. Deep learning for image super-resolution: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(10):3365–3387, 2021. 2

[40] Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan LI, Hang Su, and Jun Zhu. Prolificdreamer: High-fidelity and diverse text-to-3d generation with variational score distillation. In *Advances in Neural Information Processing Systems*, pages 8406–8441, 2023. 3

[41] Pengxu Wei, Ziwei Xie, Hannan Lu, Zongyuan Zhan, Qixiang Ye, Wangmeng Zuo, and Liang Lin. Component divide-and-conquer for real-world image super-resolution. In *European Conference on Computer Vision (ECCV)*, pages 101–117. Springer, 2020. 5, 6

[42] Rongyuan Wu, Lingchen Sun, Zhiyuan Ma, and Lei Zhang. One-step effective diffusion network for real-world image super-resolution. In *Advances in Neural Information Processing Systems*, pages 92529–92553, 2024. 1, 3, 4, 5, 6, 8

[43] Rongyuan Wu, Tao Yang, Lingchen Sun, Zhengqiang Zhang, Shuai Li, and Lei Zhang. Seesr: Towards semantics-aware real-world image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25456–25467, 2024. 1, 3, 5, 6

[44] Zongliang Wu, Siming Zheng, Peng-Tao Jiang, and Xin Yuan. Realism control one-step diffusion for real-world image super-resolution. *arXiv preprint arXiv:2509.10122*, 2025. 2, 3

[45] Sidi Yang, Tianhe Wu, Shuwei Shi, Shanshan Lao, Yuan Gong, Mingdeng Cao, Jiahao Wang, and Yujiu Yang. Maniqa: Multi-dimension attention network for no-reference image quality assessment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 1191–1200, 2022. 6

[46] Tao Yang, Rongyuan Wu, Peiran Ren, Xuansong Xie, and Lei Zhang. Pixel-aware stable diffusion for realistic image super-resolution and personalized stylization. In *European Conference on Computer Vision (ECCV)*, pages 74–91. Springer, 2024. 1, 3, 6

[47] Qiaosi Yi, Shuai Li, Rongyuan Wu, Lingchen Sun, Yuhui Wu, and Lei Zhang. Fine-structure preserved real-world image super-resolution via transfer vae training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 12415–12426, 2025. 1, 3, 6

[48] Weiyi You, Mingyang Zhang, Leheng Zhang, Xingyu Zhou, Kexuan Shi, and Shuhang Gu. Consistency trajectory matching for one-step generative super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 12747–12756, 2025. 3, 6

[49] Fanghua Yu, Jinjin Gu, Zheyuan Li, Jinfan Hu, Xiangtao Kong, Xintao Wang, Jingwen He, Yu Qiao, and Chao Dong. Scaling up to excellence: Practicing model scaling for photo-realistic image restoration in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25669–25680, 2024. 1, 3

[50] Zongsheng Yue, Jianyi Wang, and Chen Change Loy. Resshift: Efficient diffusion model for image super-resolution by residual shifting. In *Advances in Neural Information Processing Systems*, pages 13294–13307, 2023. 3, 6

[51] Zongsheng Yue, Kang Liao, and Chen Change Loy. Arbitrary-steps image super-resolution via diffusion inversion. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 23153–23163, 2025. 3, 6

[52] Matthew D. Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In *European Conference on Computer Vision (ECCV)*, pages 818–833. Springer, 2014. 5

[53] Kai Zhang, Jingyun Liang, Luc Van Gool, and Radu Timofte. Designing a practical degradation model for deep blind image super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4791–4800, 2021. 1, 2, 6

[54] Lin Zhang, Lei Zhang, and Alan C. Bovik. A feature-enriched completely blind image quality evaluator. *IEEE Transactions on Image Processing*, 24(8):2579–2591, 2015. 6

[55] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 4, 5, 6

[56] Yuanzhi Zhu, Ruiqing Wang, Shilin Lu, Junnan Li, Hanshu Yan, and Kai Zhang. Oftsr: One-step flow for image super-resolution with tunable fidelity-realism trade-offs. *arXiv preprint arXiv:2412.09465*, 2024. 2, 3